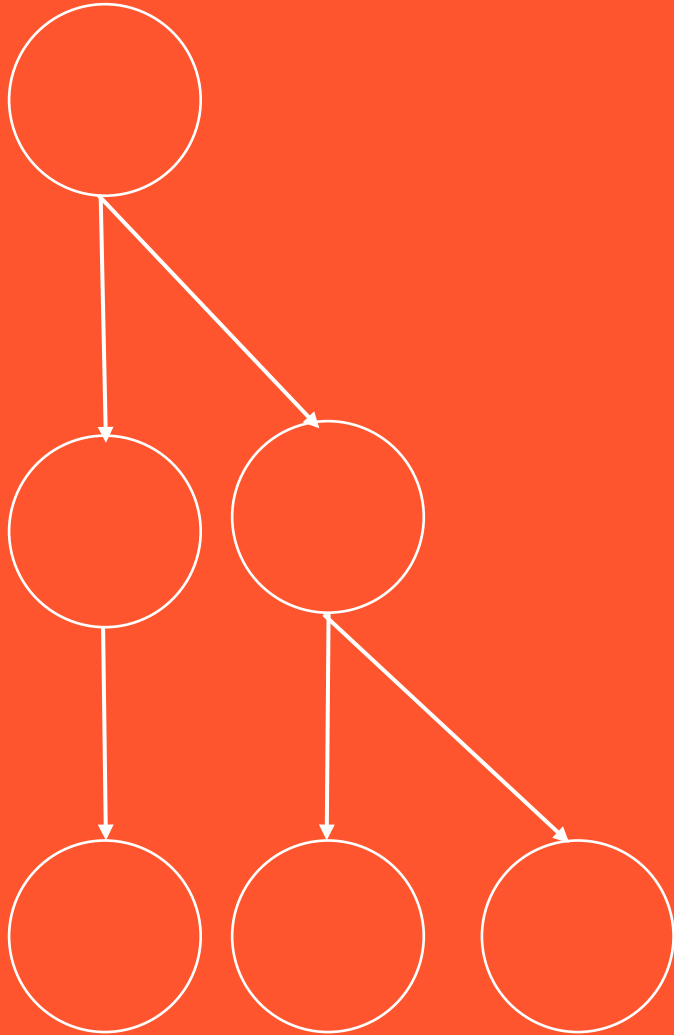


BCOG 100

Symbolic Cognition

Lecture 3: The Brain and Cognitive Science of Literacy, Math, and Science



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

Reading Check 12-3

Password: searle

Question: What is the Russian Room (or Chinese Room)
Thought experiment?

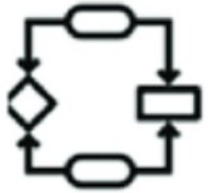


The Brain and Cog Sci of Literacy, Math, and Science



Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

Recap & Overview

Last Time:

Symbolic culture and the expressive mind: music, storytelling, and art.

Today:

Writing, numbers, science

How symbols reshape cognition

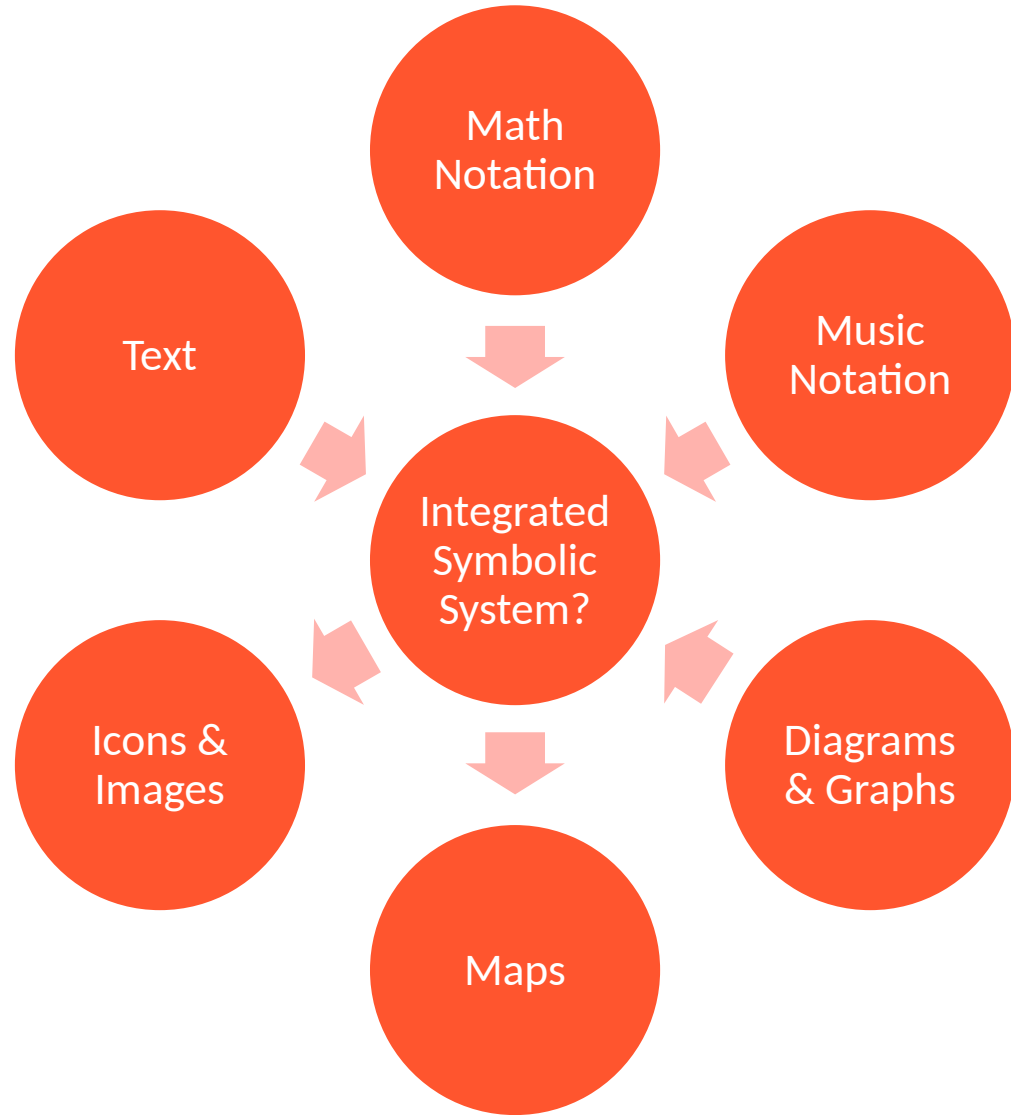
The Brain and Cog Sci of Music, Storytelling, & Art



From Symbols to Systems

- Symbols as shared units
- Systems as organized structures
- Coordination enables cumulative knowledge

The Brain and Cog Sci of Music, Storytelling, & Art



The Cognitive Demands of Symbolic Systems

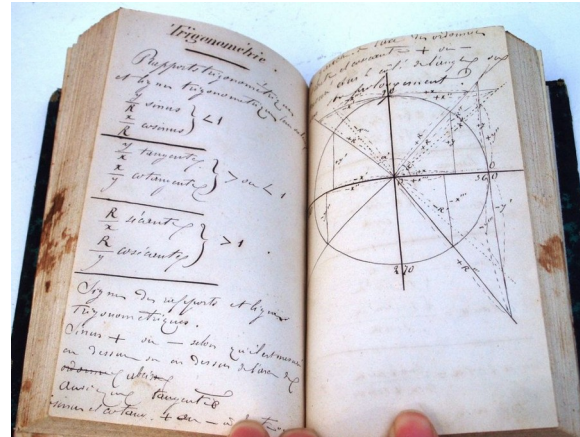
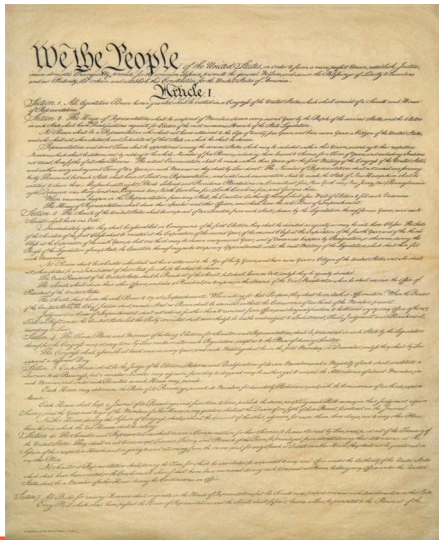
- Symbols must map to perception
- Rules must organize symbols
- Groups must share conventions

The Brain and Cog Sci of Music, Storytelling, & Art



Why Writing Appeared: The Functional Problem

- Memory limits constrain groups
- Speech disappears when spoken
- Writing preserves ideas across time



The Brain and Cog Sci of Music, Storytelling, & Art

The Cultural Evolution of Writing Systems

- Early marks tracked goods
- Scripts became more abstract
- Different cultures found different paths

Segmental (e.g., European)

<i>Alphabet</i>	ka	ki	ku	ke
<i>Syllabary</i>	か	き	く	け
<i>Abugida</i>	ക	കി	കു	കെ

Pictograms (e.g. Nisbiti)



Morphosyllabic (e.g., Chinese)

Semantic

	Sun (日)	Water (氵)
Phonetic: 青	晴 (qíng)	清 (qīng)
“qing” sound family	“sunny, clear sky”	“clear, pure”
Phonetic: 易	陽 (yáng)	湯 (tāng)
“yang” sound family	“sun rising/ brilliant sunlight”	“soup; hot water”

Japanese

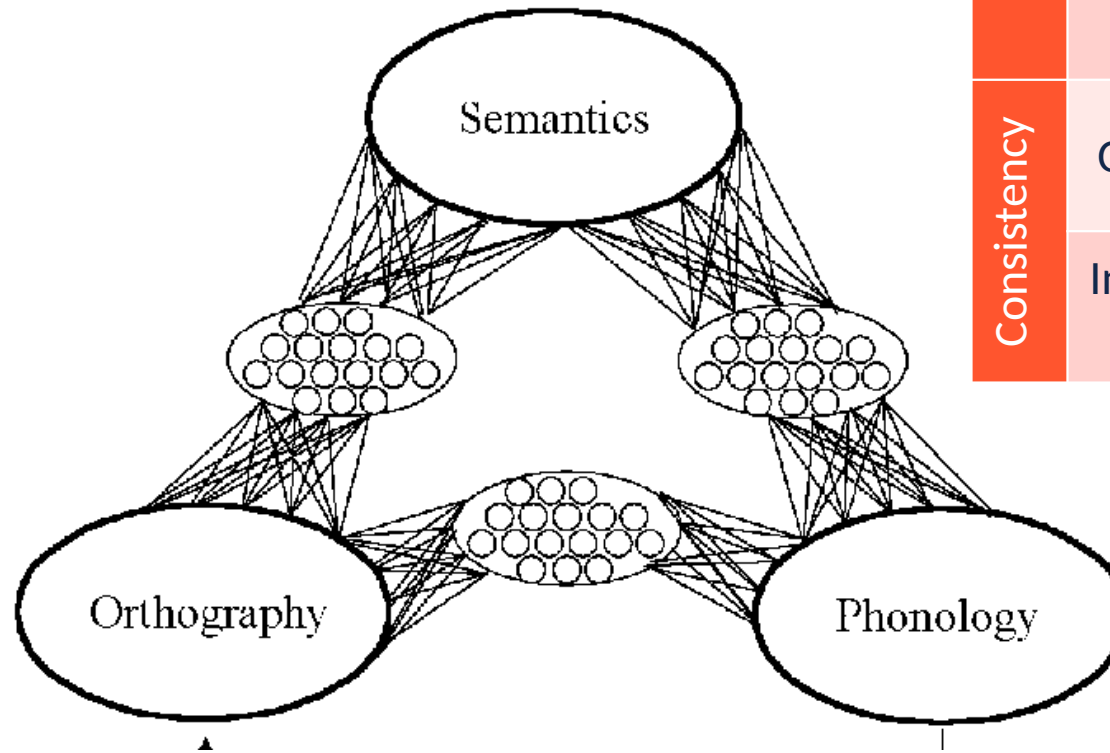
“I am an engineer.”



The Brain and Cog Sci of Music, Storytelling, & Art

The Cognitive and Neural Foundations of Reading

- Reading reuses older systems
- Visual patterns map to speech
- Networks integrate prediction and meaning

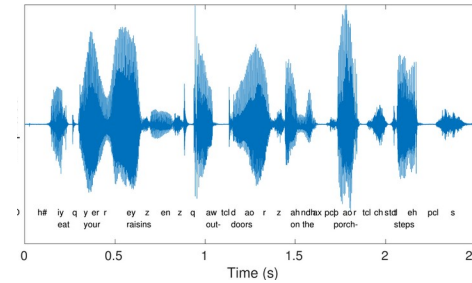


		Frequency	
		High	Low
Consistency	Consistent	SAVE	MINT
	Inconsistent	HAVE	PINT

The Brain and Cog Sci of Music, Storytelling, & Art

From Speech to Text: The Cognitive Leap

- Speech is continuous
- Writing is discrete
- Reading requires new skills



Speech: fleeting, sequential, fast, context-bound

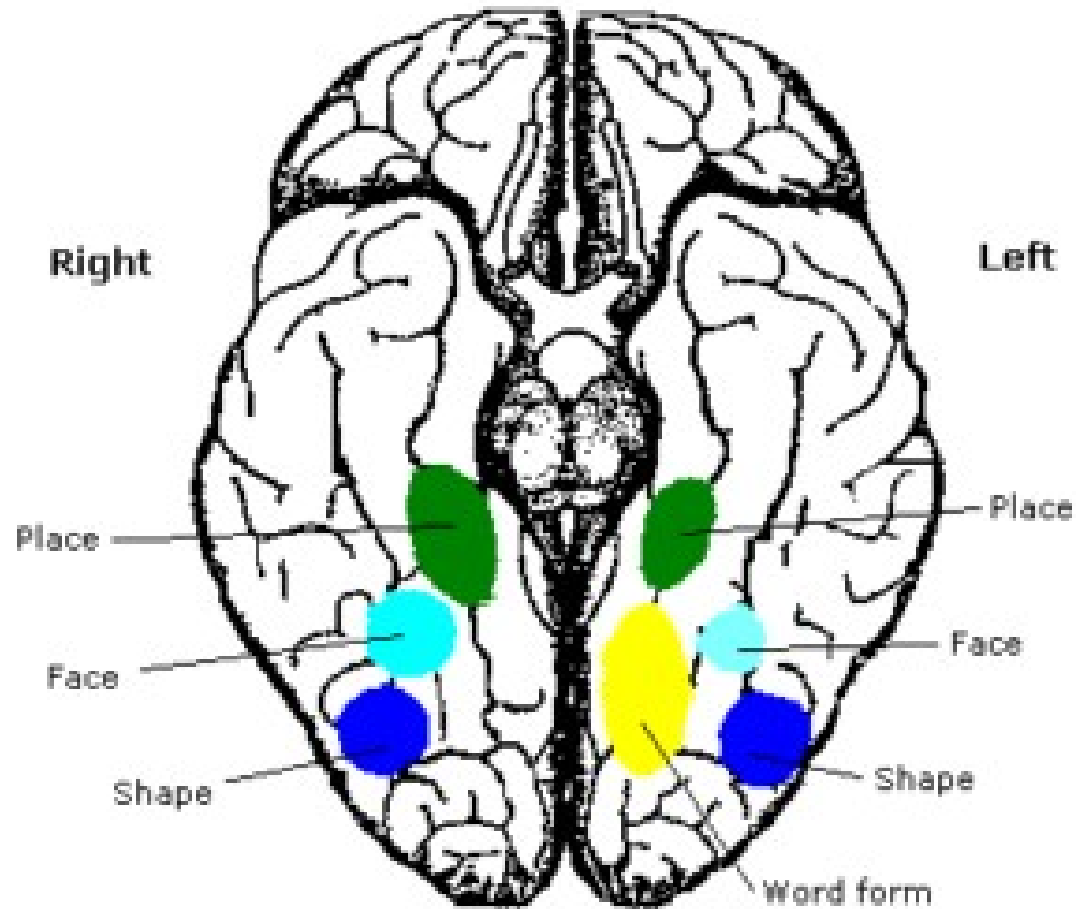


女孩很开心

Text: lasting, revisable, inspectable, transmittable across time



The Brain and Cog Sci of Music, Storytelling, & Art



How Literacy Changes the Brain

- Literacy reshapes the brain
- Different scripts recruit different systems
- Culture sculpts neural pathways

The Brain and Cog Sci of Music, Storytelling, & Art

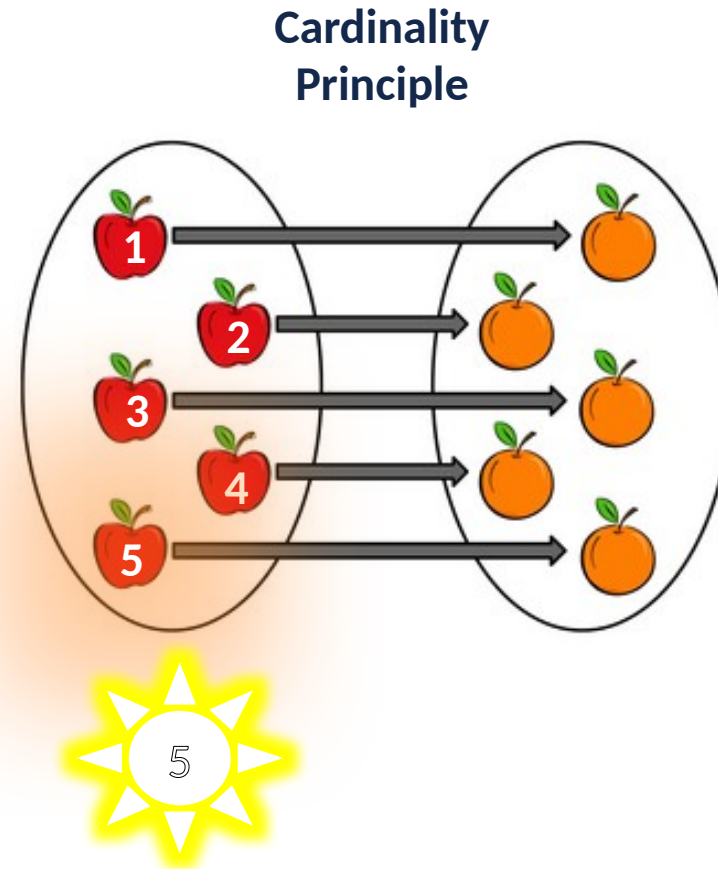
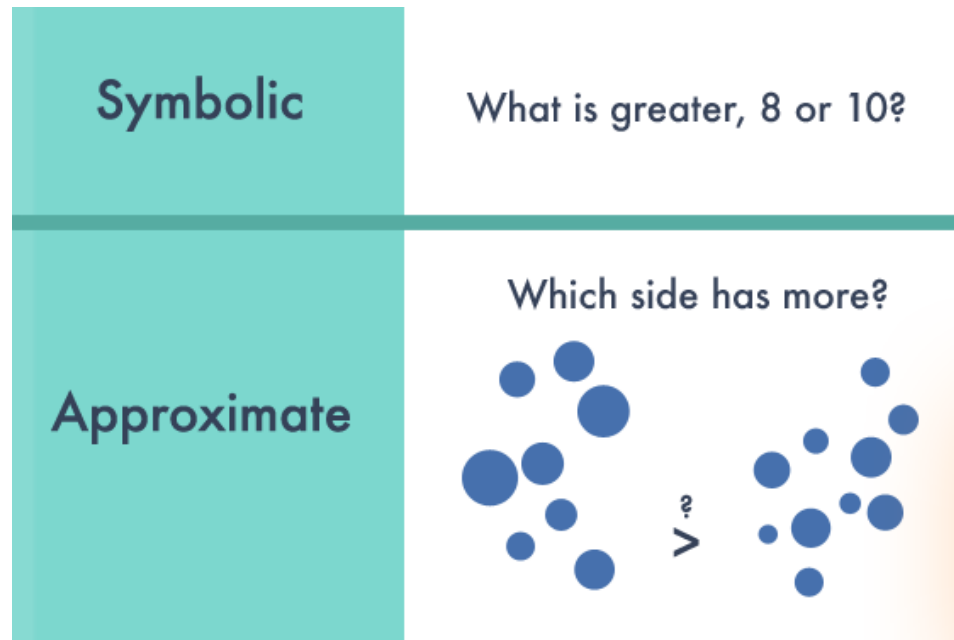
The Functional Problem:

Why Represent Numbers?

- From writing to measuring the world
- Symbols unlock new kinds of precision
- Exact number requires more than perception

Resources	Fairness	Planning
Which one is enough for the group?	How do we know the split is fair?	How many days should we stay here?

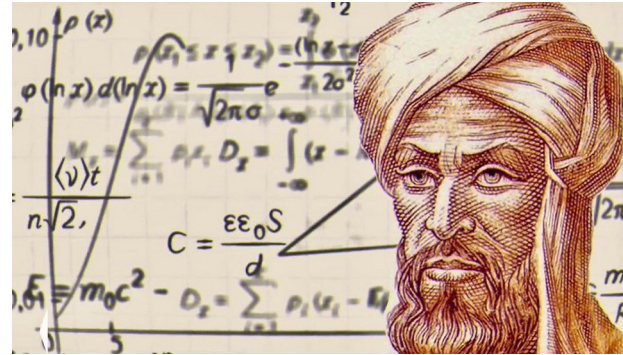
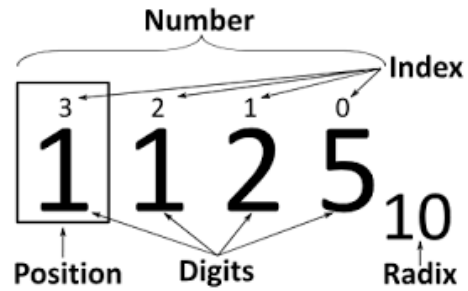
The Brain and Cog Sci of Music, Storytelling, & Art



Cognitive Origins of Number Concepts

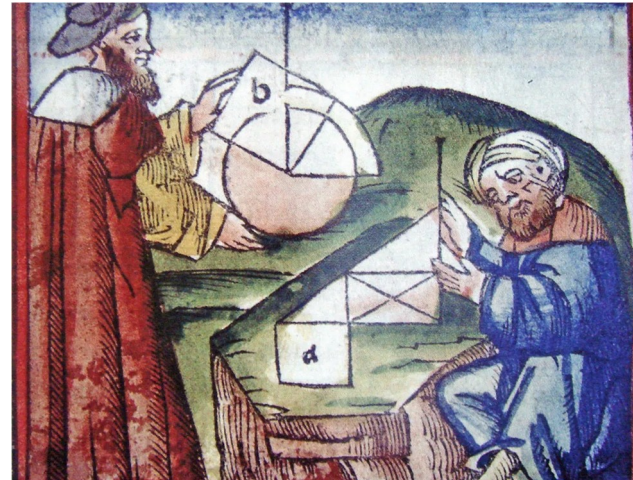
- Primate quantity systems
- From perception to symbols
- Building exact number

The Brain and Cog Sci of Music, Storytelling, & Art

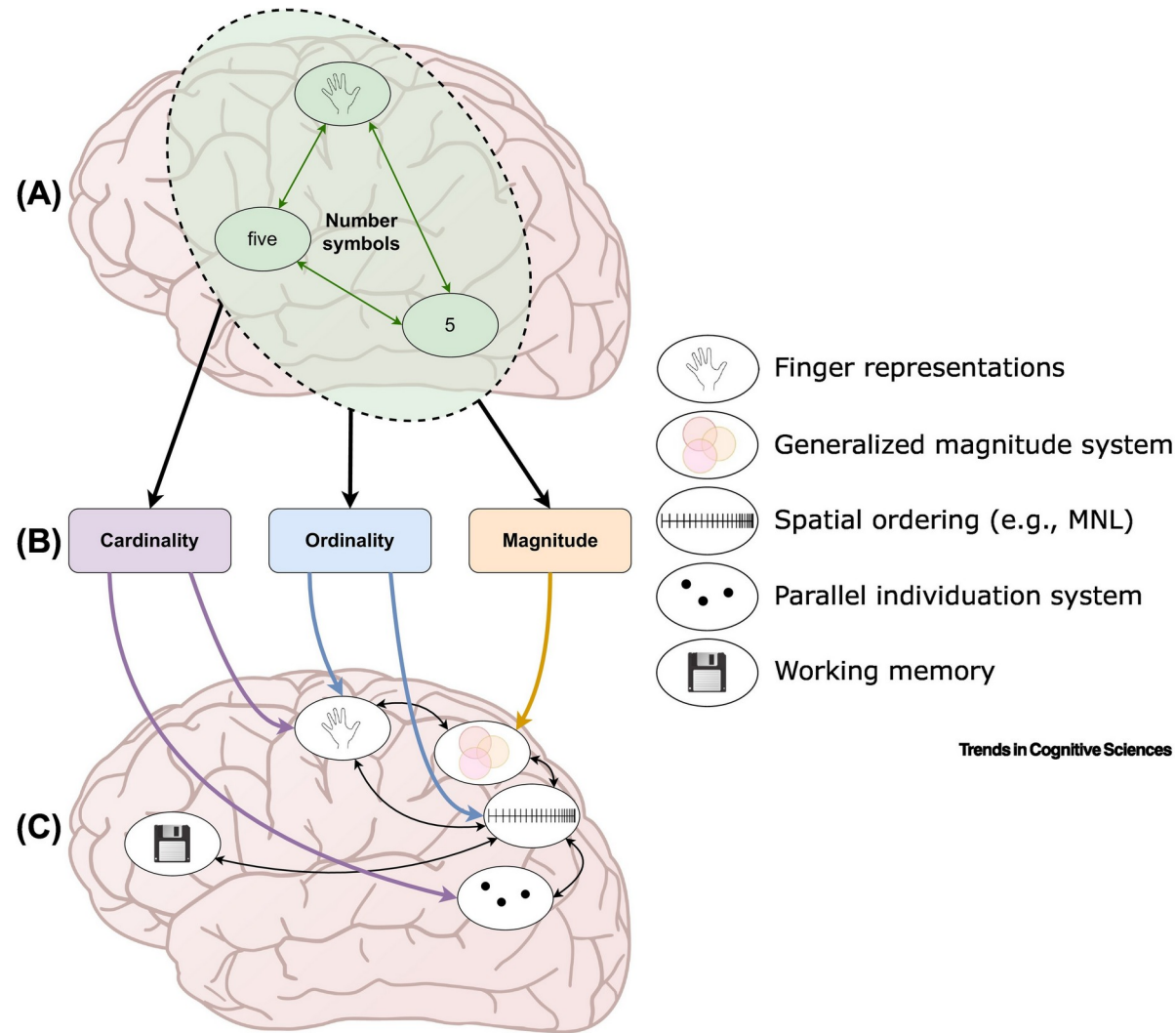


Cultural Evolution of Number Systems

- From tallies to tokens
- From words to notation
- From quantity to mathematics



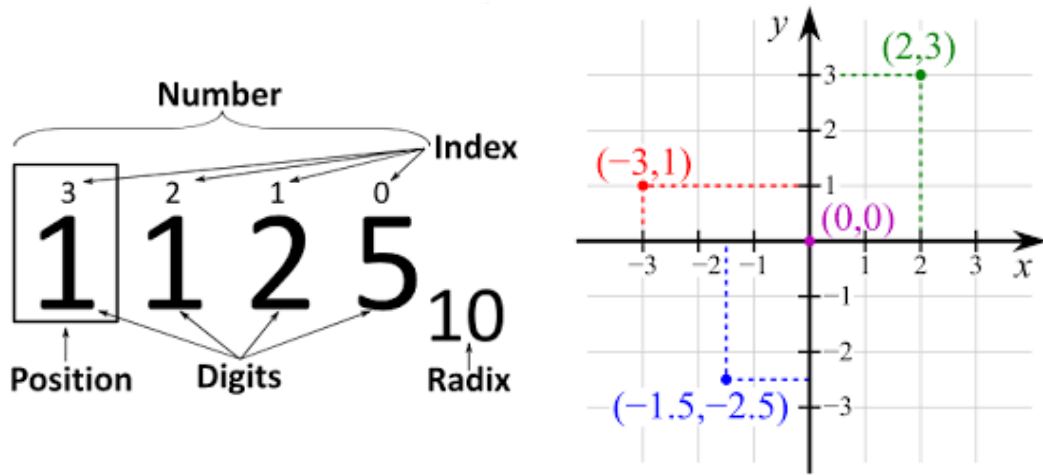
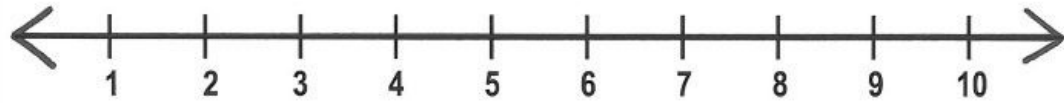
The Brain and Cog Sci of Music, Storytelling, & Art



The Mathematical Brain

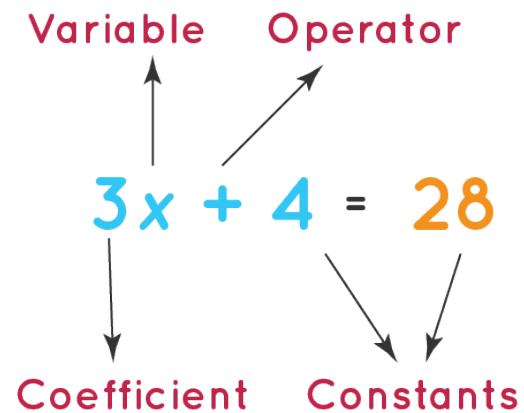
- Ancient magnitude circuits
- Learning symbolic number
- Networks for calculation

The Brain and Cog Sci of Music, Storytelling, & Art



Math as Externalized Cognition

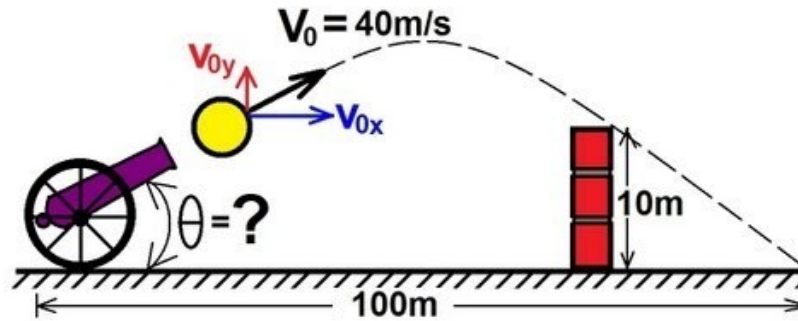
- Tools that extend the mind
- Symbols that reshape thought
- Math as cultural computation



The Brain and Cog Sci of Music, Storytelling, & Art



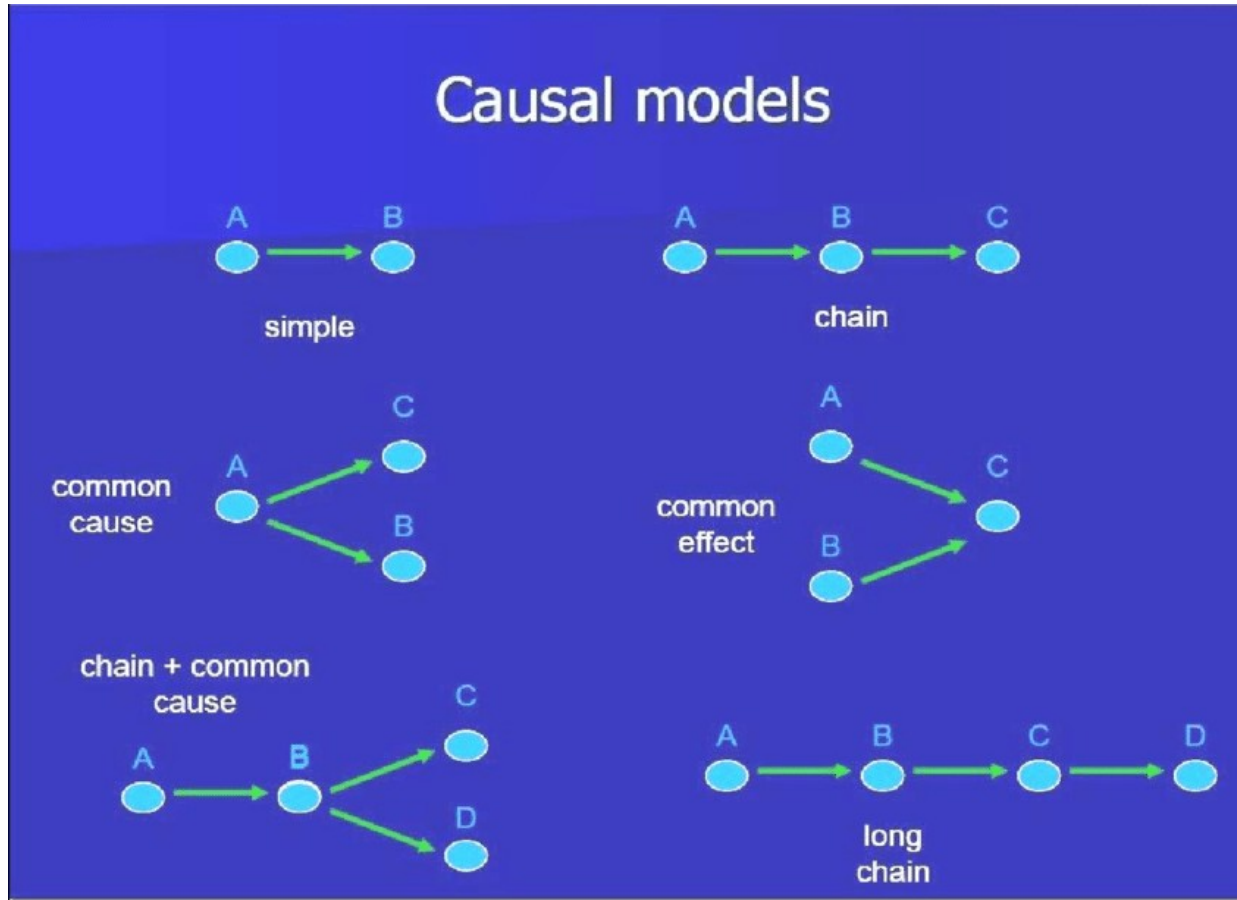
$$\begin{array}{l} t_{\text{in air}} = ? \text{ (y-direction)} \\ y = y_0 + v_{0y}t + \frac{1}{2}gt^2 \end{array} \quad \begin{array}{l} t_{\text{in air}} = ? \text{ (x-direction)} \\ x = v_{0x}t \end{array}$$



Why Science? The Functional Problem

- From words and numbers to knowledge
- Beyond intuition and memory
- A method for understanding

The Brain and Cog Sci of Music, Storytelling, & Art



Cognitive Foundations of Scientific Thinking

- Curiosity and explanation
- Models and predictions
- Limits of intuition

The Brain and Cog Sci of Music, Storytelling, & Art

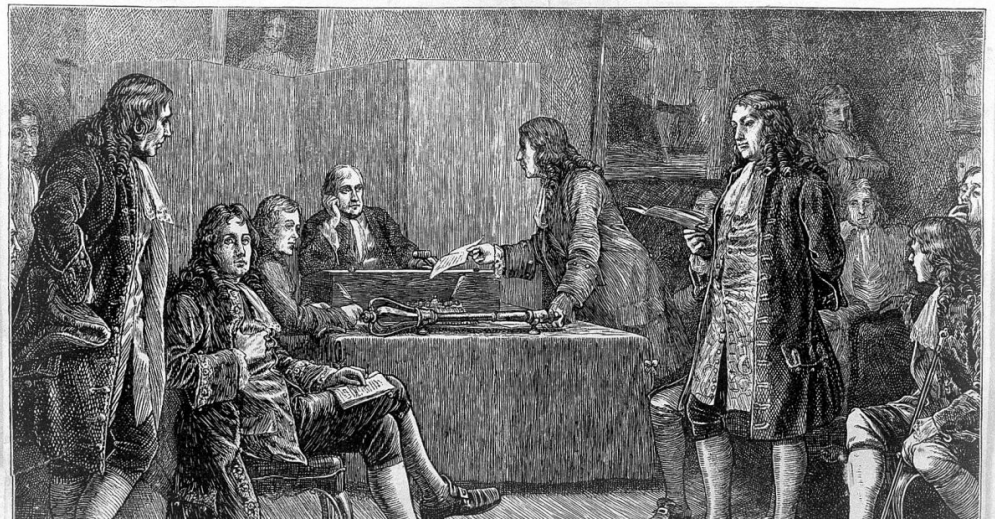
Cultural Evolution of Knowledge-Making

- Knowledge beyond individuals
- Institutions that stabilize truth
- Science as social cognition

Period	Knowledge Innovation	What It Enabled	Why It Mattered for Cumulative Culture
Prehistoric (100,000+ years ago)	Oral storytelling & teaching	Transmission of skills, norms, and myths within families or small groups	Allowed knowledge to persist across generations, but fidelity was low and memory-dependent
Early agricultural societies (10,000–5,000 BCE)	Tally marks & tokens	Tracking resources, debts, and quantities	First external memory aids; stabilized planning information
Early civilizations (3500–1500 BCE)	Writing systems (Sumerian, Egyptian, Chinese)	Records, administration, laws, historical accounts	Vastly increased fidelity and scope of knowledge transmission
Classical world (500 BCE–500 CE)	Libraries & scholarly communities	Organized repositories of texts; cross-generational comparison	Created first cumulative knowledge infrastructures
Medieval to early modern (1000–1600 CE)	Manuscript culture & commentary tradition	Preservation, reinterpretation, and refinement of earlier texts	Enabled incremental improvement of ideas over centuries
Scientific Revolution (1600–1800 CE)	Scientific societies, journals, peer review	Public evaluation, reproducibility, shared standards	Transformed knowledge-making into a collective error-correction system
Industrial & modern era (1800–2000)	Mass literacy, standardized education, printing	Wide distribution of knowledge; reproducibility at scale	Deepened the “ratchet effect” by making knowledge accessible to millions
Digital era (2000–present)	Online databases, open science, global collaboration	Real-time sharing, large data sets, meta-analysis, cross-lab replication	Dramatically accelerated cumulative culture and error correction
Emerging future	AI-assisted discovery & simulation	Automated pattern-finding, model generation, and testing	May transform the rate and structure of scientific knowledge



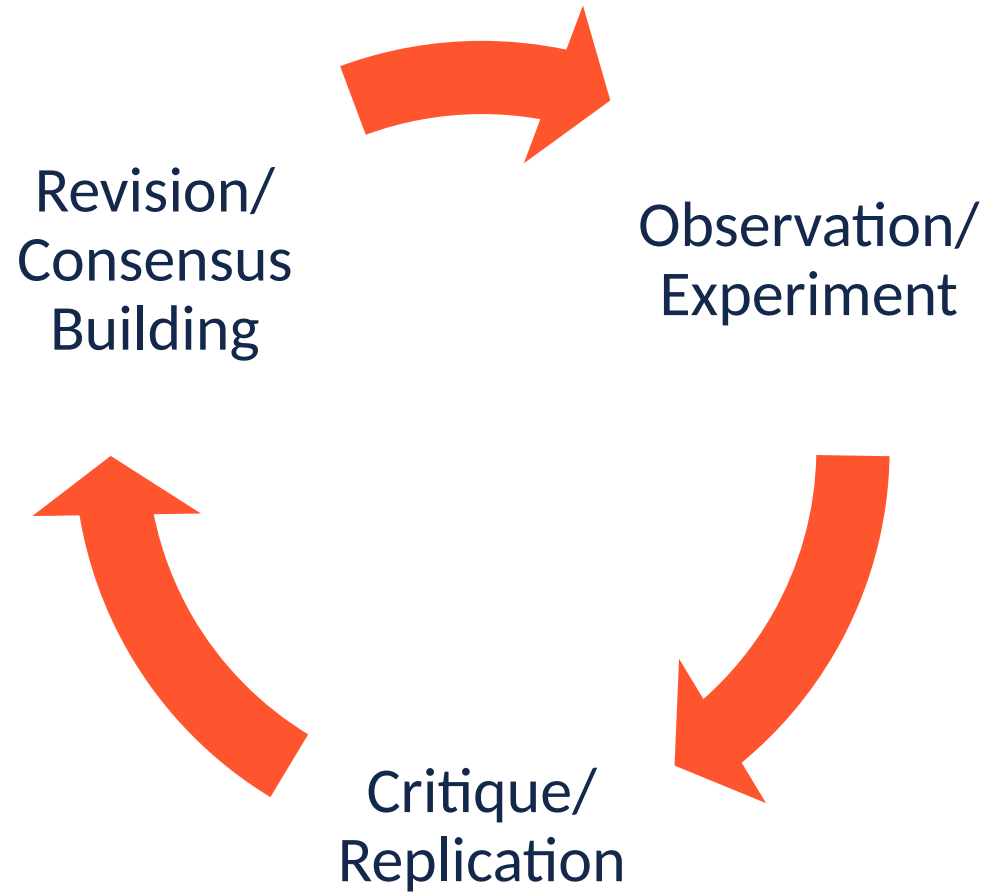
The Brain and Cog Sci of Music, Storytelling, & Art



The Social Brain and Institutional Epistemology

- Trust and cooperation
- Shared standards of evidence
- Knowledge through social networks

The Brain and Cog Sci of Music, Storytelling, & Art



Scientific Institutions as Error-Correction Machines

- From observation to methods
- From individuals to institutions
- Reforms to strengthen reliability

The Brain and Cog Sci of Music, Storytelling, & Art

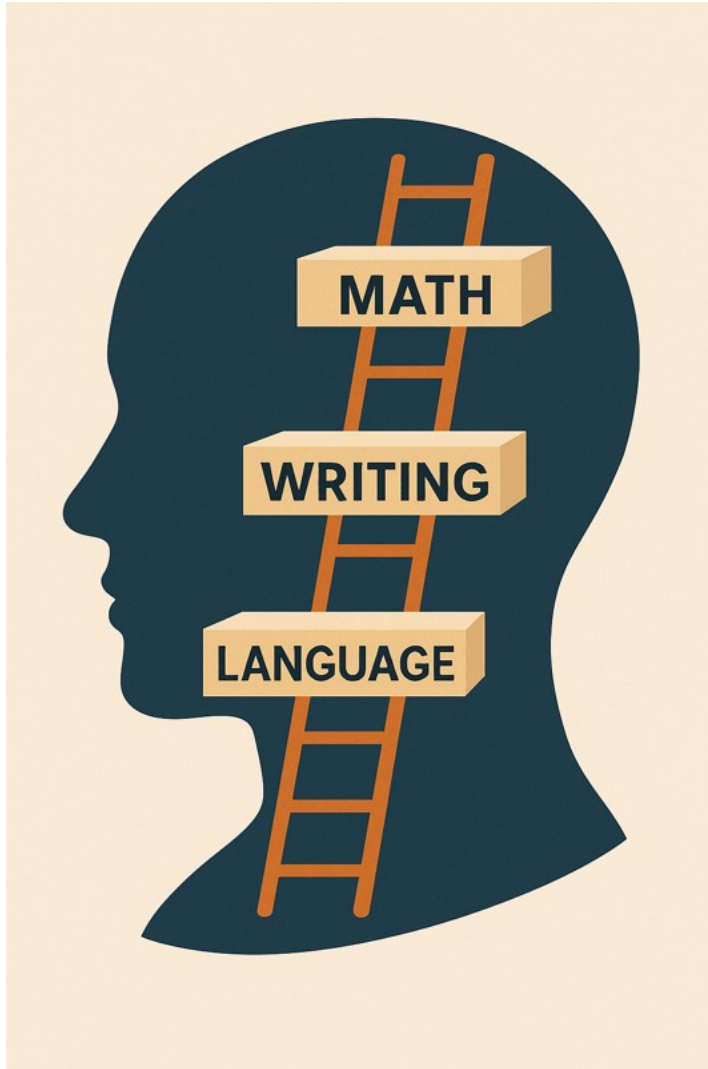
- **Frontal Cortex (PFC)** → generating hypotheses
- **Sensory Cortex** → providing evidence
- **Basal Ganglia / Dopamine System** → signaling prediction error
- **Hippocampus** → encoding new knowledge

The Neuroscience of Inquiry and Discovery

- Curiosity as a drive
- Prediction error as a guide
- Insight from recombination



The Brain and Cog Sci of Music, Storytelling, & Art



Symbolic Systems as Cognitive Amplifiers

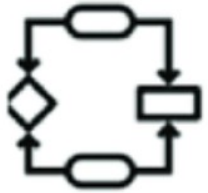
- Symbols extend cognition
- Culture stabilizes knowledge
- Brains, tools, and institutions coevolve

The Brain and Cog Sci of Music, Storytelling, & Art



Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

Summary

- Symbols extend the mind
- Culture amplifies intelligence
- Brains and environments co-create cognition

Next Time

- Nature ~~vs.~~ Nurture